ModelIt!

Teacher's Guide

# Fast Fashion Is Killing the Planet

Modeling the Environmental Cost of Cheap Clothing

## Quick Reference

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|---|---|
| **Grade Level** | Grade 09L1 |
| **NGSS Standard** | HS-ESS3-4, HS-ETS1-3 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# Table of Contents

ModelIt!


1 NGSS Standards Unpacking & CAST Alignment ..... 3

2 Scoring Rubric ..... 4

3 Background Content Knowledge ..... 5

4 The LEVER Framework ..... 6

5 Slide-by-Slide Facilitation Guide ..... 7

6 Answer Key & Expected Responses ..... 9

7 Collaborative Learning Protocols ..... 10

8 Differentiation & Extensions ..... 11

9 Common Misconceptions ..... 12

# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-ESS3-4, HS-ETS1-3**

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints.

## Three-Dimensional Learning

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|---|---|
| **Science Practice** | **Analyzing and Interpreting Data** — Students analyze data on water usage, carbon emissions, and waste generation across the fashion supply chain to evaluate the environmental impact of consumer behavior and production practices. |
| **Disciplinary Core Idea** | **ESS3.C: Human Impacts on Earth Systems / ETS1.B: Developing Possible Solutions** — The fashion industry demonstrates how human industrial activity impacts multiple Earth systems — water, atmosphere, and land — simultaneously. Students evaluate technological and behavioral solutions to reduce these impacts. |
| **Crosscutting Concept** | **Cause and Effect** — Students trace the causal chain from consumer purchasing decisions through industrial production to measurable environmental consequences, demonstrating how individual actions aggregate into global system-level effects. |

## CAST Testing Alignment

**ANALYZE:**

A clothing company claims it has become 'carbon neutral' by offsetting its emissions. Using your model, explain why carbon offsets alone don't address the full environmental impact of fast fashion — what other components are still causing damage?

**EVALUATE:**

Compare two solutions: (A) a new recycling technology that can process 50% of textile waste into new fabric, and (B) a cultural shift where consumers buy 50% fewer garments. Using evidence from your model, which solution would have a greater total environmental impact?

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The fashion industry has quietly become one of the most destructive industries on Earth. While we obsess over oil companies and car emissions, the clothes on your back account for 10% of global carbon emissions, consume 79 trillion liters of water annually, and generate 92 million tons of waste per year. And it's accelerating — clothing production has doubled since 2000.

## The Fast Fashion Business Model

Fast fashion is built on speed, volume, and disposability. Traditional fashion had 2-4 seasons per year; Zara introduced 24 'micro-seasons.' Shein uploads 2,000-10,000 new styles DAILY. The model works by making clothes so cheap that consumers treat them as disposable — wearing garments an average of 7 times before discarding them. The average American buys 68 garments per year, up from 25 in 1960. Globally, 150 billion garments are produced annually for 8 billion people.

## Water: The Hidden Cost

Cotton is the most water-intensive crop used in fashion. Growing enough cotton for a single T-shirt requires 2,700 liters of water — equivalent to one person's drinking water for 2.5 years. Dyeing and finishing add more: textile dyeing is the second-largest polluter of clean water globally. The Aral Sea, once the world's fourth-largest lake, has virtually disappeared due in part to cotton irrigation for the textile industry. The fashion industry consumes 79 trillion liters of water annually.

## Carbon Emissions: Fashion's Climate Impact

The fashion industry produces approximately 1.2 billion tons of CO2 equivalent per year — 10% of global emissions. This exceeds international flights and maritime shipping combined. The carbon comes from every stage: petroleum-based synthetic fibers (polyester = plastic = oil), energy-intensive manufacturing, global shipping, and methane from decomposing textile waste in landfills. If the industry continues on its current trajectory, its emissions will increase by 50% by 2030.

## Textile Waste: Mountains of Clothes

Of the 150 billion garments produced annually, 87% end up in landfills or incinerators. Less than 1% of clothing is recycled into new clothing. Even donated clothes face grim fates — only 10-20% of donated items are sold domestically; the rest are exported to developing nations where they're often dumped in landfills, destroying local textile industries. In Ghana's Kantamanto market, 40% of imported secondhand clothes arrive in such poor condition they go directly to the dump.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify consumer demand, production rate, textile waste, water consumption, and carbon emissions as the key components of the fast fashion environmental impact system.

**E** **Establish Relationships**
Students determine that consumer demand drives production rate, which simultaneously increases textile waste, water consumption, and carbon emissions — a one-directional cascade with demand as the primary driver.

**V** **Visualize & Model**
Students build a computational model showing how purchasing decisions cascade through the supply chain to create measurable environmental damage across multiple Earth systems.

**E** **Evaluate Outcomes**
Students run business-as-usual, buy-less, and circular fashion scenarios to compare the environmental impact of demand reduction versus technological solutions.

**R** **Revise & Extend**
Students add microplastic pollution, labor exploitation, or chemical runoff to model more complete environmental and social impacts of the fashion industry.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Look at your outfit right now. How much water did it take to make? How much CO2? Where will it end up when you're done with it? The answers might shock you."*

**Watch For**

Have students look at their clothing labels. Ask who knows where their clothes were made. Show map of global garment production.

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we're following your T-shirt from cotton field to landfill — and measuring the environmental destruction at every step."*

**Watch For**

Have students read objectives. Pre-teach 'water footprint' and 'carbon footprint.' Share the 2,700 liters per T-shirt fact.

## Slide 3: Big Question

3 min

**Say:** *"You paid five dollars. But the planet paid 2,700 liters of water, 7 kilograms of CO2, and a spot in a landfill for the next 200 years. That's the real price tag."*

**Watch For**

Show the true cost infographic. Quick-write: Why do you think the environmental cost isn't included in the price?

## Slide 4: LEVER Framework

2 min

**Say:** *"We're going to model the system that turns your shopping habit into an environmental disaster — and then figure out how to fix it."*

**Watch For**

Preview LEVER steps. Emphasize that this is a SYSTEM — changing one thing changes everything.

### Slide 5: Activity 1: Components

8 min

**Say:** *"What's the one thing that drives this entire system? And what are the environmental consequences?"*

**Watch For**

Guide sorting: consumer demand is external (it's the input WE control). Production rate, waste, water, and carbon are internal system responses.

### Slide 6: Activity 2: Connections

8 min

**Say:** *"When you click 'add to cart,' what happens next? Trace the impact from your phone to the cotton field to the landfill."*

**Watch For**

Help students map the complete supply chain. Identify how demand drives production, which drives ALL environmental outputs.

### Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let's test three futures: keep shopping like we do, buy half as much, or recycle everything. Which one actually saves the planet?"*

**Watch For**

Students predict outcomes. Key discovery: reducing demand beats recycling because most damage happens during production, not disposal.

### Slide 8: Discoveries

5 min

**Say:** *"The most sustainable garment is the one you already own. Recycling helps — but buying less helps MORE."*

**Watch For**

Lead discussion on why demand reduction is more effective than recycling. Connect to social media's role in driving trend cycles.

### Slide 9: STEM Challenge

12 min

**Say:** *"A billion-dollar industry wants to go green without losing teenage customers. Design their solution."*

**Watch For**

Groups design sustainable fashion solutions. Must be appealing to teen market AND reduce environmental impact. Present business pitches.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Consumer Demand
- **INTERNAL:** Production Rate, Textile Waste, Water Consumption, Carbon Emissions

## Relationships

- **Consumer Demand to Production Rate:** POSITIVE (+) — Higher consumer demand signals manufacturers to increase production speed and volume. Fast fashion brands use real-time sales data and social media trends to ramp up production of popular styles within days.
- **Production Rate to Textile Waste:** POSITIVE (+) — Higher production rates generate more manufacturing waste (fabric scraps, defective items, overstock) and more consumer waste as garments are treated as disposable. Approximately 12% of fabric is wasted during cutting alone.
- **Production Rate to Water Consumption:** POSITIVE (+) — More garments produced means more cotton grown (irrigated), more fabric dyed (water-intensive), and more textiles washed and finished. Water consumption scales almost linearly with production volume.
- **Production Rate to Carbon Emissions:** POSITIVE (+) — Increased production requires more energy for manufacturing, more petroleum for synthetic fibers, more fuel for global shipping, and generates more landfill methane from unsold and discarded garments.

## Key Discoveries

- Consumer demand is the primary driver — every additional garment purchased triggers water consumption, carbon emissions, and eventual textile waste in a one-directional cascade
- The fast fashion system produces environmental damage at every stage: growing raw materials (water), manufacturing (carbon), shipping (carbon), wearing (microplastics), and disposal (landfill waste)
- Reducing demand has a larger environmental impact than improving recycling, because most environmental damage occurs during production, not disposal
- The fashion industry's environmental impact is hidden from consumers — the true cost of a $5 T-shirt includes 2,700 liters of water, 7 kg of $CO_2$, and eventual landfill space for 200+ years

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Donating old clothes is sustainable" | Only 10-20% of donated clothing is sold domestically. The rest is exported to developing nations, where up to 40% ends up in landfills — the same fate it would have had at home, just in a poorer country. Donation makes us FEEL sustainable without actually solving the waste problem. The most impactful action is buying less in the first place. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Recycled clothing solves the waste problem" | Less than 1% of clothing is recycled into new clothing. Most 'recycled' textiles are downcycled into insulation, rags, or stuffing — and even that represents a tiny fraction of total waste. The technology to turn an old T-shirt into a new T-shirt at scale doesn't exist yet. Recycling is important but cannot offset the volume of waste created by current production levels. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Sustainable fashion is too expensive for normal people" | Fast fashion is artificially cheap because environmental costs are externalized — someone else pays for the water, carbon, and waste. A $5 shirt isn't actually cheap; the true cost is just hidden. Moreover, cost-per-wear analysis shows that a $50 quality garment worn 100 times ($0.50/wear) is cheaper than a $5 garment worn 5 times ($1.00/wear) before falling apart. |